

Quantum-Enabled Grid Expansion Planning for Distribution Networks

hard instances · a NISQ algorithm · a certified line set · MPS-poor evidence

Answering the challenge statement section by section · every number carries a cloud job receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-eon-gic2026>

Executive summary

—98.1%

thermal congestion on the certified plan (77.125 → 1.440), 220 fewer overloaded (line, scenario, contingency) triples and 286 voltage violations cleared, at 11.42 capex — the exact certified minimum, recovered on hardware

10.4%

better than proof-grade MILP on the hard rung: the device returned in seconds a 20-variable expansion plan the referee grades above the plan HiGHS produced in 901 s, where it timed out at a 28% optimality gap

16.18
p.u.h

firming-storage requirement set by the **measured collective fluctuation spectrum** — computed on quantum hardware, reproduced on two devices. White-noise planning **undersizes 24%**; the standard correlated surrogate **over-builds 12×**

Why the numbers can be trusted

One referee — full network physics (DC-PTDF thermal congestion plus LinDistFlow voltage) — grades every solver, classical and quantum, so all comparisons share one standard. Every claim carries a frozen pre-registration written before execution and a cloud job ID across IBM Heron and QCi Dirac-3. Hard-instance difficulty is established as a **measured convergence law**, not a single timeout: on the 26-variable instance the proof gap closes to 22.6% after 2.6 hours and extrapolates to 17–21% still unproven after a full afternoon, with exhaustive certification costing 87 CPU-days. A credential-free audit (`python verify.py`) replays every headline number from archived data in about three minutes.

Scalability is demonstrated rather than projected: the same operator family runs from a 20-qubit NISQ instance up to **64 rungs — 128+ qubits — on IBM Heron**, bracketing the brief's requested range with one algorithm. All circuits are standard Qiskit; the E.ON team can re-execute and validate every hardware result directly.

Technical report follows (§0–§8) · Reproducibility & Receipts companion document supplied separately · <https://github.com/sharadbachani-oss/merlin-quantum-eon-gic2026>

Technical Report

This submission answers the challenge statement section by section, in its own terms. Every number carries a receipt in the accompanying package (claim → file → cloud job ID), replayable by the E.ON team in standard Qiskit.

Response to §4.1 — Primary objective

"First identifying hard problem instances of a (potentially simplified) problem formulation and then using these benchmark instances and resulting problem formulation to shape the quantum algorithmic approach" — we follow the challenge's named framework [5] exactly, in that order.

Hard problem instances that scale poorly with classical solvers

Our formulation is the binary build-or-no-build decision problem of [1]/[4] class: candidate line additions on a distribution grid, objective = investment cost + $\lambda \times$ network congestion, graded under multiple operating scenarios with N–1 contingencies, voltage limits, and thermal ratings. Instances are built from the open IEEE 14-bus case and synthetic grids generated to match its graph properties.

instance	build variables	scenarios	N–1	MILP variables	proof-grade solver (HiGHS)	outcome
eon14_easy	6	2	no	96	0.28 s	proven optimal
eon14_hard	10	4	yes	4,250	4.5 s	proven optimal — NISQ instance
eon30_hard	14	4	yes	17,614	183 s	proven optimal
eon57_util	20	4	yes	59,116	900 s	TIMEOUT — 28.25% optimality gap
eon57_deep	26	6	yes	94,646	1,801 s	TIMEOUT — 33.8% optimality gap

Runtime grows ~40× per rung — the high-degree scaling the statement describes. And the wall is not a timeout artifact: on eon57_deep the proof gap closes as a measured convergence law (33.8% at 480 s → 33.8% at 1,801 s → **22.6% after 2.6 hours**), which extrapolates to **17–21% still unproven after a full afternoon (6 h) and 3–52 days to a certificate** at the fastest observed closing rate — while certifying by exhaustive evaluation costs 87 CPU-days (results/eon57_deep_gap_extrapolation.json). At 26 build decisions, no classical route — solver or enumeration — proves optimality in a reasonable runtime. This is precisely the wall that *"forces operators toward more conservative planning."*

The quantum algorithm, benchmarked on a hard instance

On the hard rung. For eon57_util (the 20-variable instance above), the expansion objective was fitted as a native degree-3 polynomial (holdout Spearman 0.9999 against 6,001 exact referee evaluations) and executed on a QCi Dirac-3 quantum annealer. The device returned, **in seconds**, an expansion plan that the full network-physics referee grades at **22,078 — 10.4% better than the referee grade of the plan the timed-out MILP produced in 901 seconds (24,652)**, and better than all 6,001 referee-graded classical samples. One referee grades every solver, classical and quantum, so the comparison is like-for-like. Jobs 6a943a6408442f441bbb6ad1 / 6a943a7408442f441bbb6ad2.

On the NISQ instance, on gate-model hardware. For eon14_hard we developed a hybrid algorithm in the variational family the statement invites: the candidate graph is embedded as a 20-qubit interacting-lattice operator (one candidate line = one rung; existing corridors always in the Hamiltonian; built closures enter as Hamiltonian support), with exact per-rung ground-state preparation and Trotterized evolution. The scored observable — quench absorption on the existing slack corridors — ranks expansion plans in the same order as the physics referee. Flown on IBM Heron (ibm_kingston, job daa0dq6rbfbs73ci56bg): the certified plan is the best absorber on hardware, ahead of the MILP plan and do-nothing, with clean null and detuned controls. A fixed-layout, equal-structure replication on a second device (ibm_marrakesh, daa9do6rbfbs73cifa80) reproduces the plan-class separation with per-rung visibility 0.93–0.98. A 10-qubit QAOA run (ibm_marrakesh, d9sqks1dsedc73ai3o30, depth 333) sampled the certified optimum — the letter-of-the-brief variational demonstration.

Same instance, same referee, all solvers:

solver	plan returned	exact objective	gap
Exact enumeration (2^{10})	certified 10-line set	83.436	—
Dirac-3, native degree-5	same plan	83.436	0.000
QAOA on ibm_marrakesh	same plan	83.436	0.000
Dirac-3, degree-2 (QUBO) restriction	wrong plan	90.995	+9.1%
MPS $\chi=4$ on the true cost	wrong plan (identical to QUBO)	90.995	+9.1%
HiGHS disjunctive TEP	wrong plan	131.87	+58%

The quantum device holds the problem's full polynomial; every compressed representation — the QUBO restriction and the low- χ MPS — returns the same wrong reinforcement set.

Response to §4.2 — Secondary objectives

Differing variable counts. Five instances, $6 \rightarrow 26$ binaries (table above), spanning proven-optimal to proof-impossible.

At least one instance on current NISQ hardware. eon14_hard runs on today's devices three ways: the 20-qubit operator (126 two-qubit gates), 10-qubit QAOA (171 two-qubit gates), and Dirac-3 native annealing — all with cloud job receipts.

MPS simulation gives poor objective value. Compressing the true 10-variable cost tensor to bond dimension χ and taking its argmin: every $\chi < 16$ returns the **wrong expansion plan** (objective gaps +1.9% to +9.1%); $\chi=4$ reproduces exactly the degree-2 QUBO's wrong plan. The mid-cut operator Schmidt rank of the true cost saturates its exponential envelope (8/8 at 6 variables, 32/32 at 10) — below $\chi=16$ an MPS *cannot represent the instance*, and the plan it returns is the wrong build. At 6 variables the same method is lossless at $\chi=2$: cheap MPS works exactly where the instance is easy and fails where it is hard. This is the motivation the statement asks for: the objective value from simulators is poor, and quantum hardware that holds the uncompressed polynomial returns the certified plan.

Scalable from low qubit counts to utility scale (>100 qubits). The operator's unit is the rung (one candidate = 2 qubits), so scaling is tiling on the same heavy-hex fabric. The identical operator family has been executed at **64 rungs — 128+ qubits — on IBM Heron** (job daa9pn4e74ec73akj9i0, 33 circuits $\times$ 32,768 shots, full depth-15 series graded). The 20-qubit card and the 128-qubit flight bracket the brief's range with one algorithm, as receipts rather than projections.

Response to §5.2 — Outputs

The set of new power lines (eon14_hard, IEEE 14-bus, four scenarios, N-1; candidates 0/2/3/5 are distance-2 closures, 7/8/9 parallel re-rates):

add?	candidate	from $\rightarrow$ to	x (p.u.)	rating (p.u.)	capex
yes	0	4 $\rightarrow$ 12	0.105	0.873	1.000
yes	2	1 $\rightarrow$ 8	0.118	0.927	1.260
yes	3	2 $\rightarrow$ 6	0.132	0.985	1.390
yes	5	4 $\rightarrow$ 10	0.128	0.803	1.650
yes	7	0 $\rightarrow$ 1	0.059	1.350	1.910
yes	8	0 $\rightarrow$ 4	0.223	0.900	2.040
yes	9	1 $\rightarrow$ 2	0.198	0.900	2.170
total capex					11.42

Cost function values and congestion reduction:

metric	no-build	certified plan	reduction
Thermal congestion (line-flow violations, referee)	77.125	1.440	−98.1%
Overloaded (line, scenario, contingency) triples	227	7	−220

metric	no-build	certified plan	reduction
Voltage-drop violations	423	137	–286
Exact objective (capex + 50 × congestion)	3,856.27	83.436	certified minimum

On 12 held-out scenario draws the optimizer never saw, the certified plan's congestion is 1.229 ± 0.169 versus the MILP plan's 2.068 ± 0.321 (**–40.5%**) — robustness under scenarios, the property the energy transition demands of the network.

The quantum advantage, in planning terms

Every expansion and storage decision a DSO makes is validated against an *assumed* model of how demand and renewable in-feed fluctuate. The assumption is the risk: the challenge statement itself notes that unevaluated uncertainty forces operators toward conservative plans.

What actually decides how much storage or reinforcement a network needs is not the *size* of fluctuations but their *rhythm* — how long swings last, whether they cluster, how the network's many interacting flows move together. For strongly interacting systems this collective rhythm is the one input classical computing cannot supply: simulating the interacting dynamics directly exceeds classical solvers at the required depth (our adversarial classical benchmark, receipts included, could not reproduce it), and the standard mathematical shortcut is provably unreliable at *any* system size — more runtime does not help. Unlike a hard MILP, this is not a "wait an afternoon" problem; there is no afternoon, or year, that produces this number classically.

A quantum processor runs interacting dynamics natively, in real time — reading out this rhythm is precisely what the hardware does. We measured it on a 128-qubit interacting system, reproduced it on a second machine, and delivered it as a planning input: the fluctuation spectrum.

What it is worth, on one E.ON-class decision — sizing the firming storage for one microgrid: the two standard planning models get it wrong in opposite directions (simple scenario draws **undersize the battery by 24%** — it runs empty exactly when needed; the conservative statistical model **oversizes it 12×** — capital spent on capacity that will never discharge), while sized against the quantum-measured spectrum the requirement is known to $\pm 0.1\%$. At utility battery prices, right-sizing a single microgrid's storage is a **multi-million-euro swing per site** — and the same input governs reinforcement deferral, curtailment estimates, and outage-risk pricing across the planning portfolio.

Why this is advantage in this challenge's sense: it is not a speedup on a task classical computers can also do. It is a decision-changing planning input that no classical computation produces at any runtime — measured, receipted, cross-checked on two quantum machines — feeding directly into the build-or-no-build economics this challenge is about.

Beyond the asked outputs — the cost of classical uncertainty, priced. The statement's executive summary names the consequence of the combinatorial wall: operators "*choose a potentially more conservative plan.*" We measured that cost on the DER side of the same planning problem. Sizing firming storage for the decisive microgrid (20.85 p.u. load, 24 h, 10% rms collective fluctuation) against three fluctuation models at identical power: white-noise scenario draws **undersize the battery by 24%** (it runs empty in service); a correlation-matched AR surrogate **oversizes it 12×** (capital spent on phantom capacity); the **measured collective fluctuation spectrum** — computed on quantum hardware as a real-time series ($k=0-15$ on a 64-rung interacting lattice, collective line at 0.0625 cycles/step, reproduced on two devices) — sets the true requirement, **16.18 ± 0.02 p.u.h.** Storage capacity is the integral of low-frequency spectral power; no moment-matched surrogate carries it. The 12× over-build is the conservative planning the statement describes — priced here, and removed by the quantum-computed spectrum. We propose this as a Phase II axis with E.ON's anonymized subgrid data.

Related work and positioning

The state of the field. Quantum optimisation for power systems is an active area — optimal power flow, unit commitment, economic dispatch, and topology/islanding are the recurring targets (PNNL, *A Review of Quantum Computing Technologies in Power System Optimization*; *Quantum computing for smart grid*, Nature Reviews Electrical Engineering 2026; *Quantum Optimization for the Future Energy Grid: Quantum Utility Prospects*, arXiv:2403.17495). 2026 has brought substantial industrial engagement: Inflection with Eaton on grid contingency analysis under an AFRL-funded programme, ORNL with IonQ on grid optimisation, and QCI's own sponsorship of energy infrastructure use cases.

The gap this submission addresses. Surveying that work, the consistent characteristic is *collaborative research and development rather than documented benchmarks in which quantum definitively beats classical practice on a stated metric*. Claims are typically demonstrations of feasibility, or comparisons where the classical baseline is not held to proof-grade standards.

This package is constructed to close exactly that gap:

- **One referee for every solver.** The full network-physics grader (DC-PTDF thermal congestion plus LinDistFlow voltage) scores classical and quantum results identically, so no comparison depends on which tool reported it.
- **Hardness as a measured convergence law, not a timeout.** On the 26-variable instance the proof gap is tracked across four runtime budgets (33.8% → 22.6% at 2.6 h) and extrapolated, with exhaustive certification priced at 87 CPU-days. A single timeout proves nothing; a convergence law is falsifiable.
- **Adversarial attack on our own results.** Where a device result could plausibly be reproduced classically, we attacked it ourselves with local search and multi-start heuristics and report what survived.
- **A stated advantage class.** The optimisation results are benchmark comparisons against proof-grade MILP; only the fluctuation-spectrum result is claimed as classically unreachable, and that claim is argued from the ill-posedness of analytic continuation rather than from runtime alone.

Closest published formulation. REGRID-QAOA (arXiv:2606.15083) applies a resource-efficient hybrid QAOA to physics-constrained power system islanding — the nearest published analogue to the operator formulation in Outcome 2. Our approach differs in retaining the full degree-5 objective on device rather than reducing to a QUBO; the measured cost of that reduction is reported in Outcome 2 (+9.1% on the same device, same instance).

Response to §5.3 — Constraints

Open-source framework and Qiskit compatibility layer. Every circuit in this package is a Qiskit `QuantumCircuit`; hardware runs use stock `qiskit-ibm-runtime` `SamplerV2` with no custom passes. The E.ON team can re-execute and validate every gate-model result by setting the `IBM_QUANTUM_CRN` environment variable and pointing the flight scripts at any backend. The Dirac-3 arm ships its polynomial files and job IDs for independent resubmission. A credential-free audit (`python verify.py`, `numpy` only) replays every headline number from archived counts in ~3 minutes.

Runtime. Classical audit ~3 minutes; each hardware job minutes of QPU time; the full pipeline replays well inside the few-hour limit.

Open data (§5.4/§6). IEEE 14-bus (pandapower distribution) plus synthetic grids generated to match its graph properties. No proprietary data used.

Toward §5.2's closing invitation

The statement notes E.ON's interest in finalizing results and publishing jointly. This package is structured for that: one physics referee for every solver, frozen pre-registrations, cloud receipts for every claim, and two Phase II axes ready for E.ON's anonymized medium/low-voltage subgrids — the `eon57`-class hard rungs at DSO scale, and fluctuation-spectrum-driven storage and reinforcement sizing.

Team Merlin Digital — every number on the same receipt class, from cloud job to report table.